Kai Barker

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EDUCATION

University of California, Santa Barbara
Statistics and Data Science B.S.
Major GPA: 3.86/4.0
Relevant Coursework: Machine Learning, Time Series, Stochastic Processes, Advanced Statistical Models, Computational Science, Differential Equations, Technology Management, Probability and Statistics, Problem Solving in Computer Science
Clubs: Consult Your Community, Statistics and Data Science Club, Data Science Collaborative, Ski and Snow Club

WORK EXPERIENCE

Consult Your Community October 2025- Present
Business Analyst Santa Barbara, CA

- Produce business readiness assessment for a small business preparing for acquisition, analyzing operations, financials, and marketing to identify streamlining opportunities and reduce owner dependency
- Analyze financials, marketing strategies, and operational workflows to increase projected revenue by $40,000
- Design 10+ Tableau visualizations and data-driven SOP to strengthen operational and financial systems

Department of Statistics and Applied Probability September 2025- Present
Undergraduate Researcher Santa Barbara, CA

- Analyze 70,000+ Senate stock trades from 2012-2024 using Python, identify statistical significance in excess returns
- Calculate risk-adjusted performance using Sharpe ratios, finding members achieved 1.97 vs. benchmark of 0.79
- Develop data cleaning pipeline to validate trade reporting accuracy and filter incomplete disclosures

Foundations of Financial Technology Lab September 2024- Present
Research Assistant Santa Barbara, CA

- Examined 20M+ blocks, analyzing wealth distribution across 100M+ wallets, calculating Gini coefficient of 0.96
- Produced comprehensive visualizations and statistical analysis tracking inequality evolution over a 5 year period
- Filtered institutional and Sybil addresses to reveal top 1% of wallets control 95%+ of Ether supply

PROJECTS

NeurIPS Ariel Data Challenge July 2024- September 2025
Competition Santa Barbara, CA

- Developed ensemble machine learning models (XGBoost + Gaussian Process Regression + Ridge) to predict exoplanet atmospheric compositions from 200+ GB of noisy spectral telescope data for ESA's Ariel mission
- Preprocessed and analyzed 14,000 complex image files across 1,100+ exoplanets, implementing advanced signal processing and noise reduction techniques for the observations
- Achieved a GLL score of 0.311 predicting atmospheric signals and uncertainties across 283 wavelength channels

Statistical Modelling for NBA Player Performance July 2024- October 2024
Personal Project Santa Barbara, CA

- Developed machine learning models, engineered features to predict player rebounds using time series NBA API data
- Implemented SHAP for model interpretability, significantly improved prediction accuracy over baselines by 20%
- Processed and analyzed performance outcomes to identify trends and validate predictions against real outcomes

SKILLS AND INTERESTS

Programming: Python, R, SQL, JavaScript, Java, C++, CSS, HTML, Typescript

Technical Skills: Microsoft Excel, Tableau, Feature Engineering, Statistical Modelling, Data Visualization, Data Pipeline Development, Git, API Integration, Canva, PowerPoint, Pandas, NumPy, Matplotlib, Scikit-learn

Achievements: HS Valedictorian, Seal of Biliteracy (Spanish), UCSB Dean’s List for Academic Excellence

Interests: Basketball, Hiking, Camping, Backpacking, Cooking, Running, Rock Climbing, Pickleball